

*Same as well
Just better*

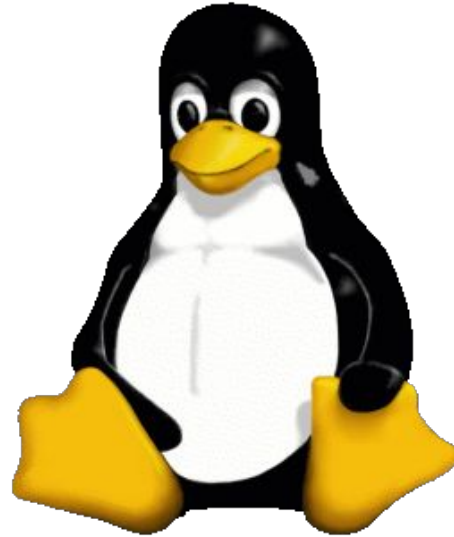
SAME YOU SINCE 2009
JUST BETTER

Before we start,

Let's recap on the
homework assignment

Lesson 2

Advanced Linux & Bash Scripting



git

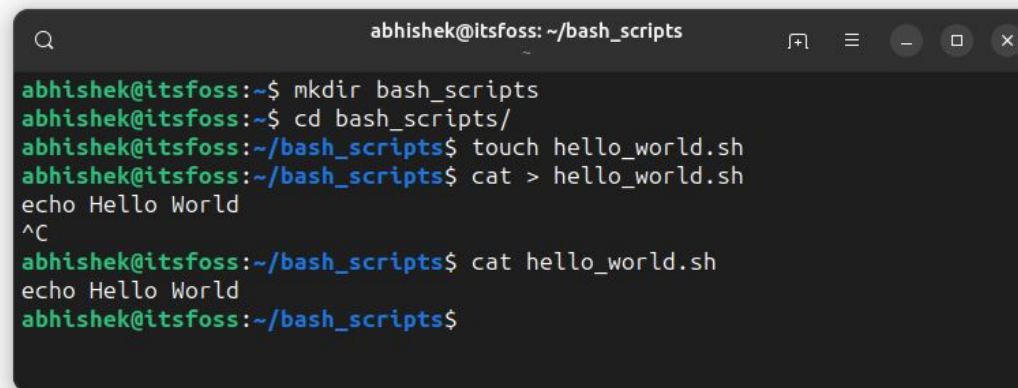
Linus Torvalds: Creator of Linux & Git.

His philosophy: Write code to solve your own annoyances.

The Problem: Typing commands manually is slow and leads to human error.

The Solution: Bash Scripting.

Key Concept: "If you do it twice, automate it." Bash is the glue of the Linux server.



```
abhishek@itsfoss: ~/bash_scripts
abhishek@itsfoss:~$ mkdir bash_scripts
abhishek@itsfoss:~$ cd bash_scripts/
abhishek@itsfoss:~/bash_scripts$ touch hello_world.sh
abhishek@itsfoss:~/bash_scripts$ cat > hello_world.sh
echo Hello World
^C
abhishek@itsfoss:~/bash_scripts$ cat hello_world.sh
echo Hello World
abhishek@itsfoss:~/bash_scripts$
```

The Shebang (`#!/bin/bash`): Tells the OS which interpreter to use.

Variables (`USER="Arman"`): Storing data to keep code clean.

Comments `#`: Giving commands some context.

Execution (`./script.sh`): Running your masterpiece.

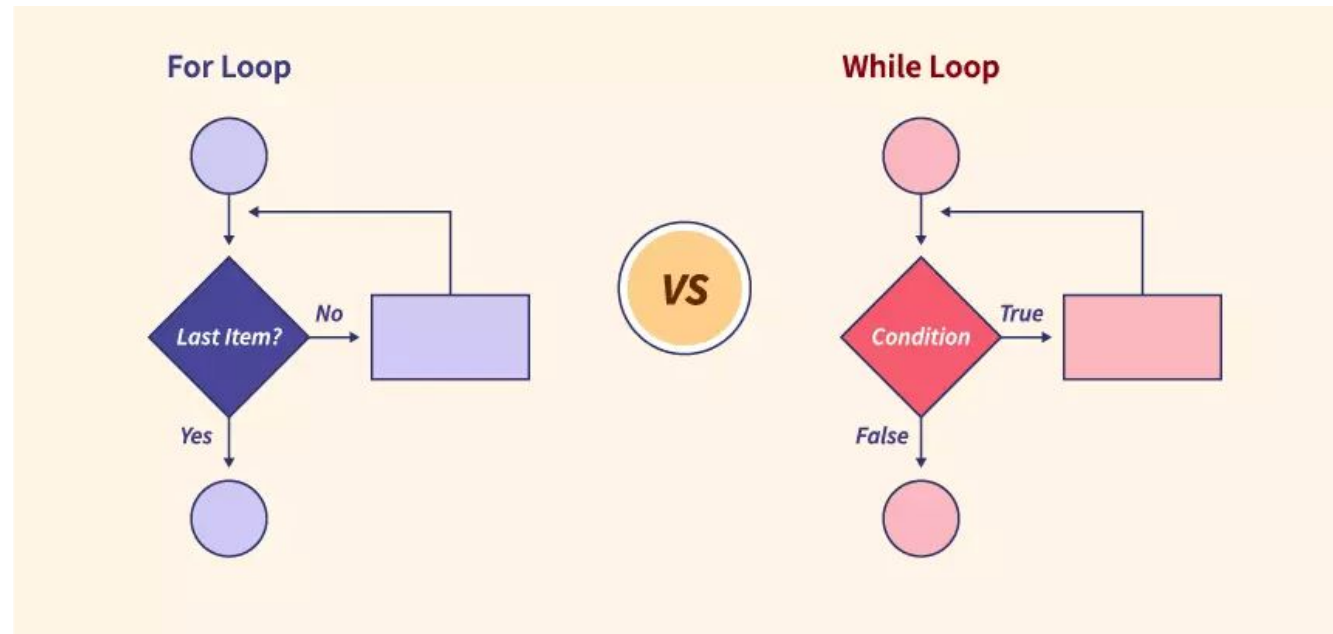
Control Flow Statements: Programming constructs for decision-making and repetition, such as if/else statements, for loops, and while loops.

Command-line Arguments: Scripts can accept input when run, accessible via special variables like `$1`, `$2`, etc.



for loops: "Do this action to every file in this folder."

while loops: "Keep doing this action until I say stop."





chmod: Read (4) | Write (2) | eXecute (1)

chown: Changing the owner of a file.

sudo: Borrowing the ultimate power (root).

```
~ (0.025s)
chmod -R g+X tmp

~ (0.025s)
ls -lR tmp

total 0
drwxr-xr--@ 3 razvan  staff  96 Jan 31 15:15 dir_1
-rw-----@ 1 razvan  staff   0 Jan 31 15:15 script.sh

tmp/dir_1:
total 0
-rwx--x---@ 1 razvan  staff   0 Jan 31 15:15 script_1.sh
```

```
ubuntu@246bf5f46b50:~ (0.032s)
find /var/www -type d -exec sudo chown johndoe:developers {} \;

ubuntu@246bf5f46b50:~ (0.012s)
ls -lR /var/www

/var/www:
total 4
drwxr-xr-x 3 johndoe developers 4096 Feb 10 12:52 dir_1
-rw-r--r-- 1 ubuntu  root         0 Feb 10 12:52 script.js

/var/www/dir_1:
total 4
drwxr-xr-x 2 johndoe developers 4096 Feb 10 12:52 dir_2
-rw-r--r-- 1 ubuntu  root         0 Feb 10 12:52 script_2.js

/var/www/dir_1/dir_2:
total 0
-rw-r--r-- 1 ubuntu root 0 Feb 10 12:52 script.sh

ubuntu@246bf5f46b50:~
```



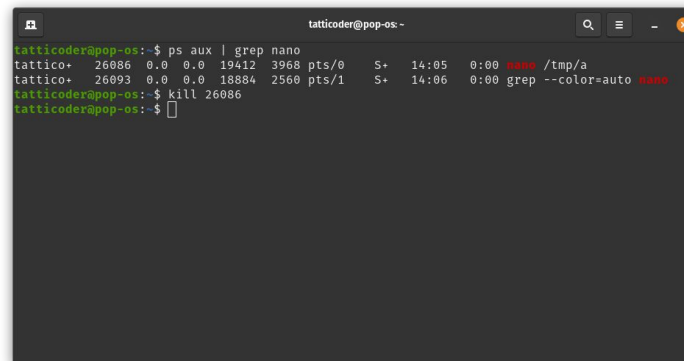

Process Management (systemd)

Daemons: Programs running silently in the background.

systemctl: Start, stop, and enable services.

journalctl: Reading the server's logs to find out why it crashed.

kill <PID>: Terminating a frozen process.



```
tatticoder@pop-os:~$ ps aux | grep nano
tattico- 26086  0.0  0.0 19412 3968 pts/0    S+   14:05   0:00 nano /tmp/a
tattico- 26093  0.0  0.0 18884 2560 pts/1    S+   14:06   0:00 grep --color=auto nano
tatticoder@pop-os:~$ kill 26086
tatticoder@pop-os:~$
```

Now, it's time for the Quiz



Challenge: Fix a broken VM using a single automated Bash script.

- Instructions will be posted in Discord.

Questions?

CONTACT US